PRODIGY_DS_05

March 19, 2026

[76]:
```
import pandas as pd
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings("ignore")
```

[77]:
```
df = pd.read_csv('US_Accidents_March23.csv')
df
```

[77]:
```
                 ID   Source  Severity           Start_Time  \
0              A-1  Source2         3  2016-02-08 05:46:00
1              A-2  Source2         2  2016-02-08 06:07:59
2              A-3  Source2         2  2016-02-08 06:49:27
3              A-4  Source2         3  2016-02-08 07:23:34
4              A-5  Source2         2  2016-02-08 07:39:07
…            …      …       …                  …
7728389  A-7777757  Source1         2  2019-08-23 18:03:25
7728390  A-7777758  Source1         2  2019-08-23 19:11:30
7728391  A-7777759  Source1         2  2019-08-23 19:00:21
7728392  A-7777760  Source1         2  2019-08-23 19:00:21
7728393  A-7777761  Source1         2  2019-08-23 18:52:06

                     End_Time  Start_Lat   Start_Lng   End_Lat    End_Lng  \
0        2016-02-08 11:00:00  39.865147  -84.058723       NaN        NaN
1        2016-02-08 06:37:59  39.928059  -82.831184       NaN        NaN
2        2016-02-08 07:19:27  39.063148  -84.032608       NaN        NaN
3        2016-02-08 07:53:34  39.747753  -84.205582       NaN        NaN
4        2016-02-08 08:09:07  39.627781  -84.188354       NaN        NaN
…                       …        …          …       …         …
7728389  2019-08-23 18:32:01  34.002480 -117.379360  33.99888 -117.37094
7728390  2019-08-23 19:38:23  32.766960 -117.148060  32.76555 -117.15363
7728391  2019-08-23 19:28:49  33.775450 -117.847790  33.77740 -117.85727
7728392  2019-08-23 19:29:42  33.992460 -118.403020  33.98311 -118.39565
7728393  2019-08-23 19:21:31  34.133930 -117.230920  34.13736 -117.23934

         Distance(mi)  … Roundabout Station   Stop Traffic_Calming  \
0               0.010  …      False   False  False           False
1               0.010  …      False   False  False           False
2               0.010  …      False   False  False           False
```

```
3                0.010  …      False   False  False           False
4                0.010  …      False   False  False           False
…                …  …         …     …    …             …
7728389          0.543  …      False   False  False           False
7728390          0.338  …      False   False  False           False
7728391          0.561  …      False   False  False           False
7728392          0.772  …      False   False  False           False
7728393          0.537  …      False   False  False           False

         Traffic_Signal Turning_Loop Sunrise_Sunset Civil_Twilight  \
0                 False        False          Night          Night
1                 False        False          Night          Night
2                  True        False          Night          Night
3                 False        False          Night            Day
4                  True        False            Day            Day
…                 …           …             …             …
7728389           False        False            Day            Day
7728390           False        False            Day            Day
7728391           False        False            Day            Day
7728392           False        False            Day            Day
7728393           False        False            Day            Day

         Nautical_Twilight Astronomical_Twilight
0                    Night                 Night
1                    Night                   Day
2                      Day                   Day
3                      Day                   Day
4                      Day                   Day
…                    …                     …
7728389                Day                   Day
7728390                Day                   Day
7728391                Day                   Day
7728392                Day                   Day
7728393                Day                   Day

[7728394 rows x 46 columns]
```

[78]:
```
df.info()
df.describe(include='all')
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7728394 entries, 0 to 7728393
Data columns (total 46 columns):
 #   Column                 Dtype
---  ------                 -----
 0   ID                     object
 1   Source                 object
 2   Severity               int64
```

```
 3   Start_Time             object
 4   End_Time               object
 5   Start_Lat              float64
 6   Start_Lng              float64
 7   End_Lat                float64
 8   End_Lng                float64
 9   Distance(mi)           float64
 10  Description            object
 11  Street                 object
 12  City                   object
 13  County                 object
 14  State                  object
 15  Zipcode                object
 16  Country                object
 17  Timezone               object
 18  Airport_Code           object
 19  Weather_Timestamp      object
 20  Temperature(F)         float64
 21  Wind_Chill(F)          float64
 22  Humidity(%)            float64
 23  Pressure(in)           float64
 24  Visibility(mi)         float64
 25  Wind_Direction         object
 26  Wind_Speed(mph)        float64
 27  Precipitation(in)      float64
 28  Weather_Condition      object
 29  Amenity                bool
 30  Bump                   bool
 31  Crossing               bool
 32  Give_Way               bool
 33  Junction               bool
 34  No_Exit                bool
 35  Railway                bool
 36  Roundabout             bool
 37  Station                bool
 38  Stop                   bool
 39  Traffic_Calming        bool
 40  Traffic_Signal         bool
 41  Turning_Loop           bool
 42  Sunrise_Sunset         object
 43  Civil_Twilight         object
 44  Nautical_Twilight      object
 45  Astronomical_Twilight  object
dtypes: bool(13), float64(12), int64(1), object(20)
memory usage: 2.0+ GB
```

[78]:

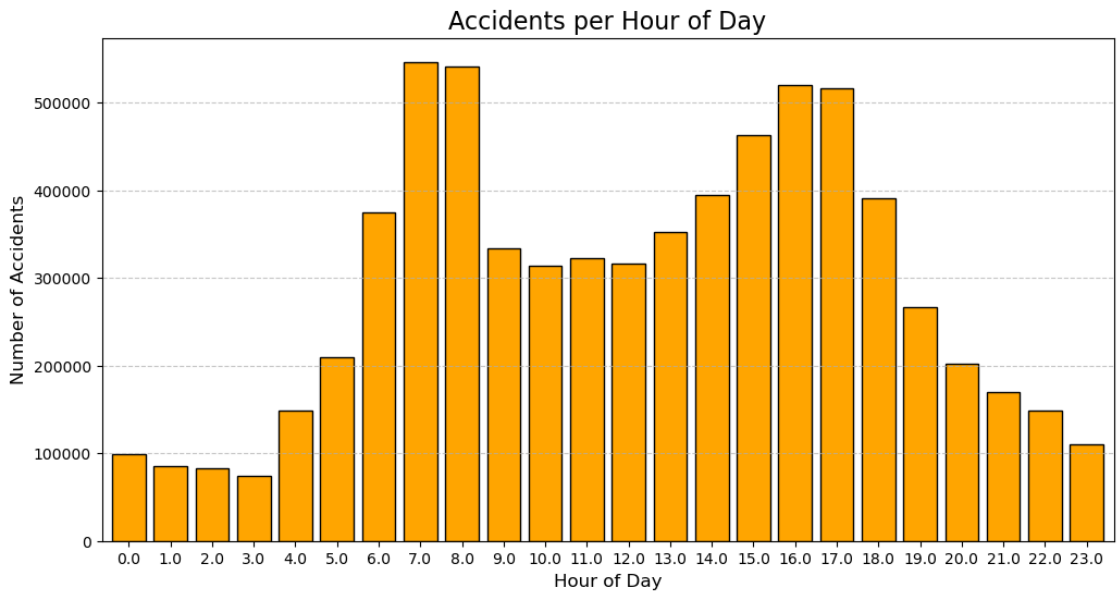
```
                ID   Source      Severity           Start_Time  \
count      7728394  7728394  7.728394e+06              7728394
unique     7728394        3           NaN              6131796
top      A-7777761  Source1           NaN  2021-01-26 16:16:13
freq           1  4325632           NaN                  225
mean          NaN      NaN  2.212384e+00                  NaN
std           NaN      NaN  4.875313e-01                  NaN
min           NaN      NaN  1.000000e+00                  NaN
25%           NaN      NaN  2.000000e+00                  NaN
50%           NaN      NaN  2.000000e+00                  NaN
75%           NaN      NaN  2.000000e+00                  NaN
max           NaN      NaN  4.000000e+00                  NaN

                   End_Time     Start_Lat     Start_Lng       End_Lat  \
count               7728394  7.728394e+06  7.728394e+06  4.325632e+06
unique              6705355           NaN           NaN           NaN
top     2021-11-22 08:00:00           NaN           NaN           NaN
freq                    112           NaN           NaN           NaN
mean                    NaN  3.620119e+01 -9.470255e+01  3.626183e+01
std                     NaN  5.076079e+00  1.739176e+01  5.272905e+00
min                     NaN  2.455480e+01 -1.246238e+02  2.456601e+01
25%                     NaN  3.339963e+01 -1.172194e+02  3.346207e+01
50%                     NaN  3.582397e+01 -8.776662e+01  3.618349e+01
75%                     NaN  4.008496e+01 -8.035368e+01  4.017892e+01
max                     NaN  4.900220e+01 -6.711317e+01  4.907500e+01

             End_Lng  Distance(mi)  … Roundabout  Station     Stop  \
count   4.325632e+06  7.728394e+06  …    7728394  7728394  7728394
unique           NaN           NaN  …          2        2        2
top              NaN           NaN  …      False    False    False
freq             NaN           NaN  …    7728145  7526493  7514023
mean   -9.572557e+01  5.618423e-01  …        NaN      NaN      NaN
std     1.810793e+01  1.776811e+00  …        NaN      NaN      NaN
min    -1.245457e+02  0.000000e+00  …        NaN      NaN      NaN
25%    -1.177543e+02  0.000000e+00  …        NaN      NaN      NaN
50%    -8.802789e+01  3.000000e-02  …        NaN      NaN      NaN
75%    -8.024709e+01  4.640000e-01  …        NaN      NaN      NaN
max    -6.710924e+01  4.417500e+02  …        NaN      NaN      NaN

       Traffic_Calming Traffic_Signal Turning_Loop Sunrise_Sunset  \
count          7728394        7728394      7728394        7705148
unique               2              2            1              2
top              False          False        False            Day
freq           7720796        6584622      7728394        5334553
mean               NaN            NaN          NaN            NaN
std                NaN            NaN          NaN            NaN
min                NaN            NaN          NaN            NaN
```

```
25%                 NaN            NaN          NaN             NaN
50%                 NaN            NaN          NaN             NaN
75%                 NaN            NaN          NaN             NaN
max                 NaN            NaN          NaN             NaN

       Civil_Twilight Nautical_Twilight Astronomical_Twilight
count         7705148           7705148               7705148
unique              2                 2                     2
top               Day               Day                   Day
freq          5695619           6076156               6377548
mean              NaN               NaN                   NaN
std               NaN               NaN                   NaN
min               NaN               NaN                   NaN
25%               NaN               NaN                   NaN
50%               NaN               NaN                   NaN
75%               NaN               NaN                   NaN
max               NaN               NaN                   NaN

[11 rows x 46 columns]
```

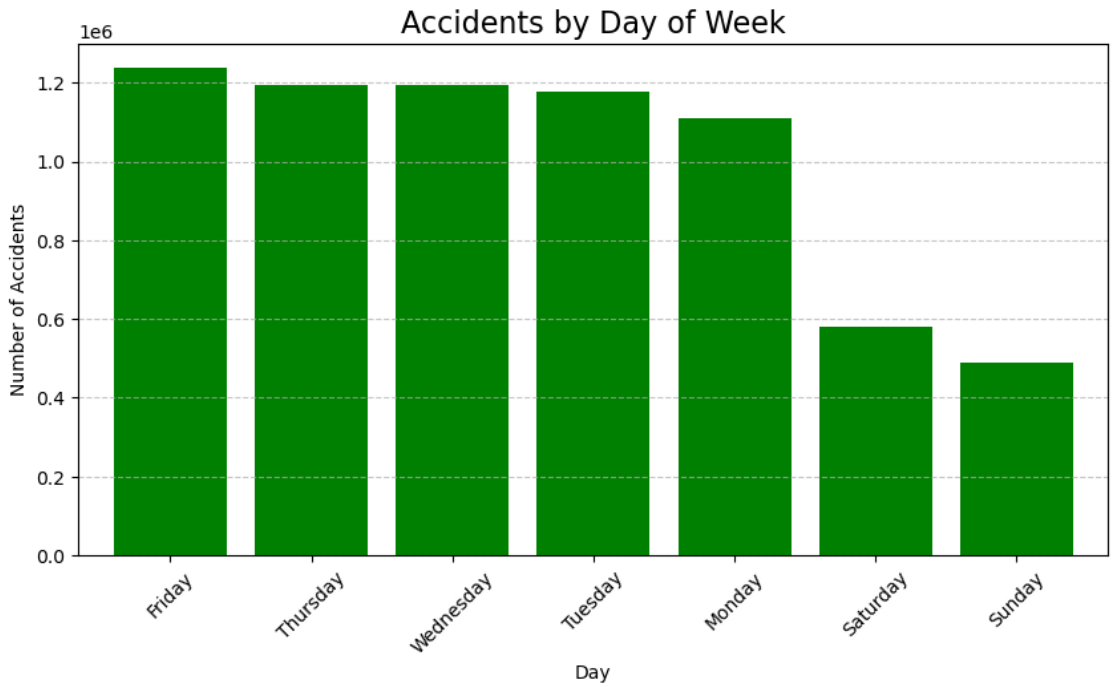
[79]:
```
# Convert datetime
df['Start_Time'] = pd.to_datetime(df['Start_Time'], errors='coerce')
df['End_Time'] = pd.to_datetime(df['End_Time'], errors='coerce')
# Extract features
df['hour'] = df['Start_Time'].dt.hour
df['day'] = df['Start_Time'].dt.day_name()
df['month'] = df['Start_Time'].dt.month
df['year'] = df['Start_Time'].dt.year

# Accident duration
df['duration'] = (df['End_Time'] - df['Start_Time']).dt.total_seconds() / 60
```

[80]:
```
plt.figure(figsize=(12,6))

df['hour'].value_counts().sort_index().plot(
    kind='bar',
    color='orange',
    edgecolor='black',
    width=0.8
)

plt.title("Accidents per Hour of Day", fontsize=16)
plt.xlabel("Hour of Day", fontsize=12)
plt.ylabel("Number of Accidents", fontsize=12)

plt.xticks(rotation=0)
plt.grid(axis='y', linestyle='--', alpha=0.7)
```

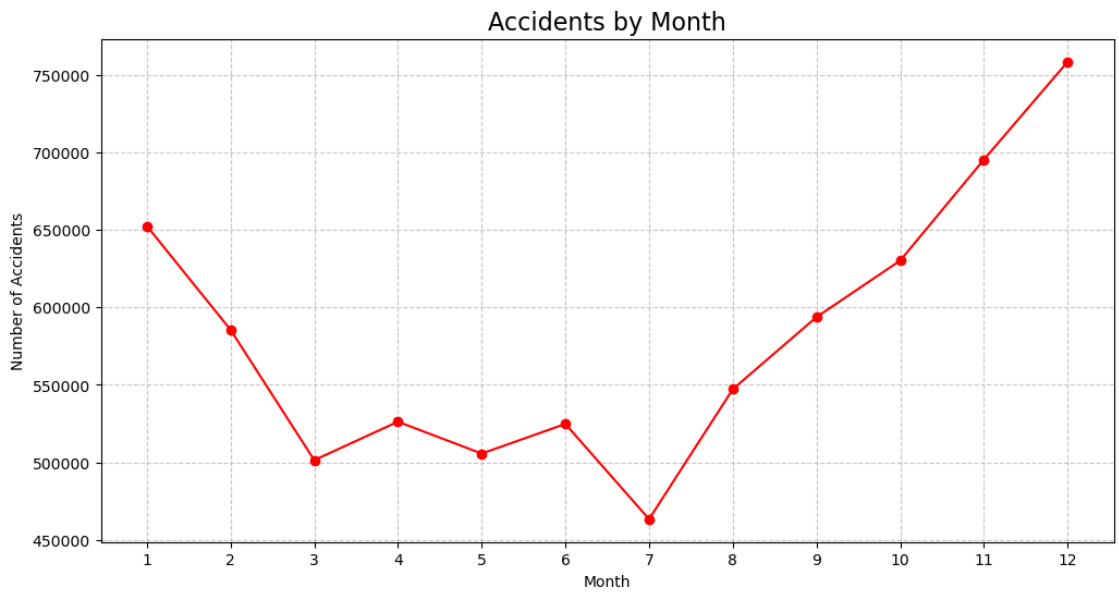
```
plt.show()
```



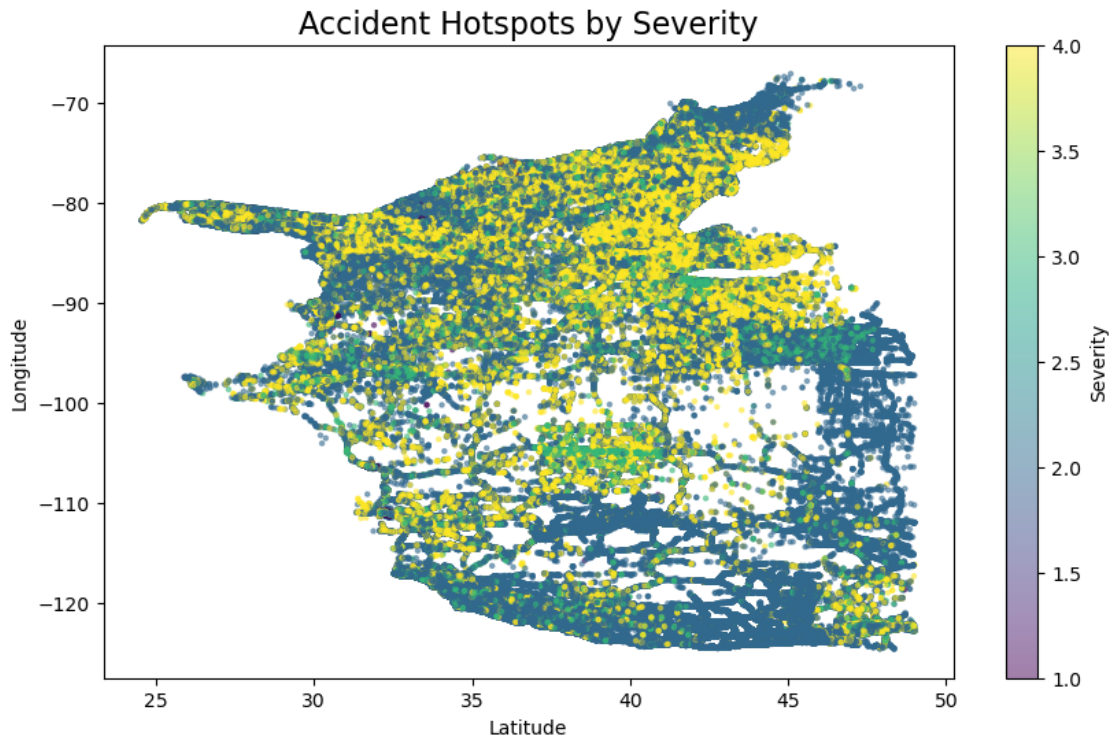
[81]:
```
plt.figure(figsize=(10,5))

df['day'].value_counts().plot(
    kind='bar',
    color = 'green',
    width=0.8
)

plt.title("Accidents by Day of Week", fontsize=16)
plt.xlabel("Day")
plt.ylabel("Number of Accidents")

plt.xticks(rotation=45)
plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.show()
```



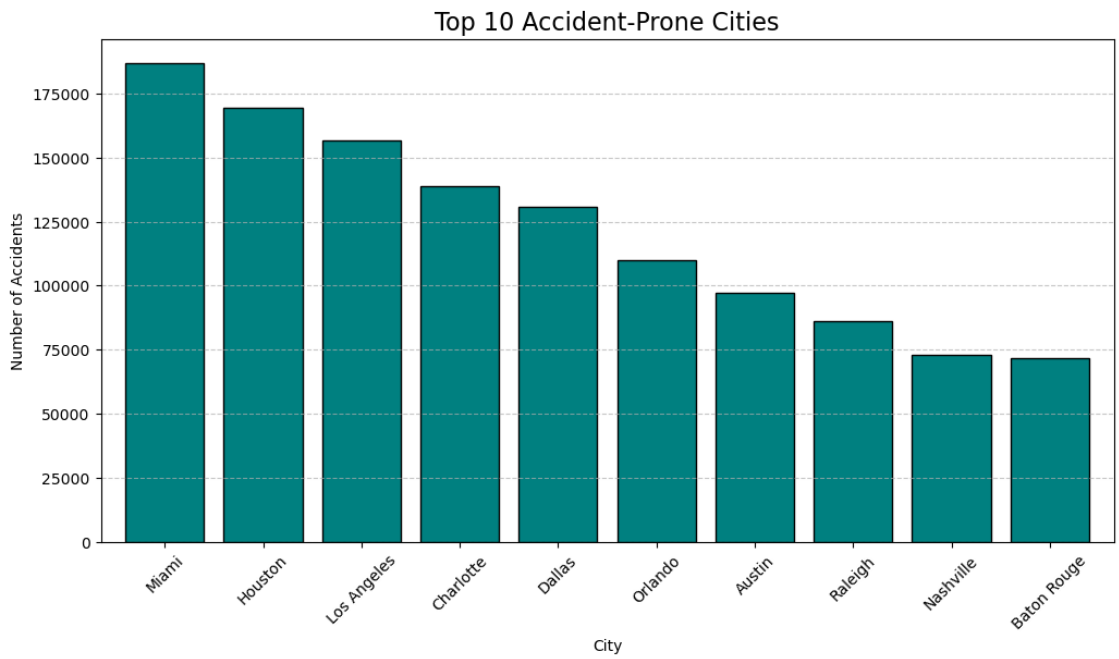
[82]:
```
plt.figure(figsize=(12,6))

df['month'].value_counts().sort_index().plot(
    kind='line',
    marker='o',
    color = 'red'
)

plt.title("Accidents by Month", fontsize=16)
plt.xlabel("Month")
plt.ylabel("Number of Accidents")

plt.xticks(range(1,13))
plt.grid(linestyle='--', alpha=0.7)

plt.show()
```



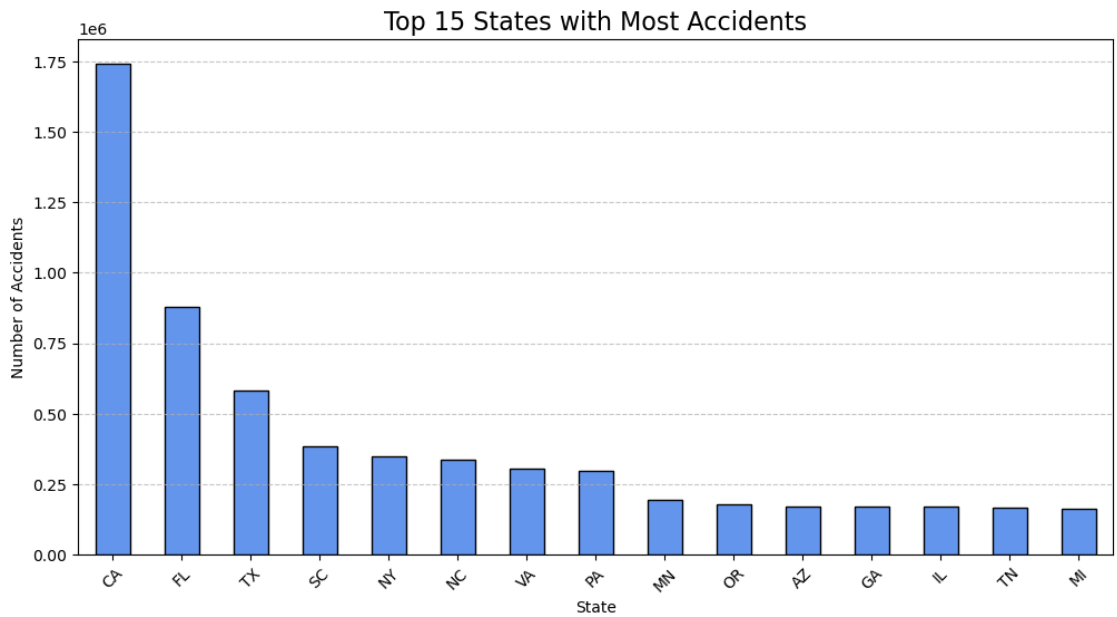
[83]:
```
plt.figure(figsize=(10,6))

plt.scatter(
    df['Start_Lat'],
    df['Start_Lng'],
    c=df['Severity'],     # color based on severity
    cmap='viridis',
    alpha=0.5,
    s=5
)

plt.colorbar(label='Severity')

plt.title("Accident Hotspots by Severity", fontsize=16)
plt.xlabel("Latitude")
plt.ylabel("Longitude")

plt.show()
```



[84]:
```
plt.figure(figsize=(12,6))

df['City'].value_counts().head(10).plot(
    kind='bar',
    color='teal',
    edgecolor='black',
    width=0.8
)

plt.title("Top 10 Accident-Prone Cities", fontsize=16)
plt.xlabel("City")
plt.ylabel("Number of Accidents")

plt.xticks(rotation=45)
plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.show()
```



[85]:
```
plt.figure(figsize=(12,6))

df['State'].value_counts().head(15).plot(
    kind='bar',
    color='cornflowerblue',
    edgecolor='black'
)

plt.title("Top 15 States with Most Accidents", fontsize=16)
plt.xlabel("State")
plt.ylabel("Number of Accidents")

plt.xticks(rotation=45)
plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.show()
```

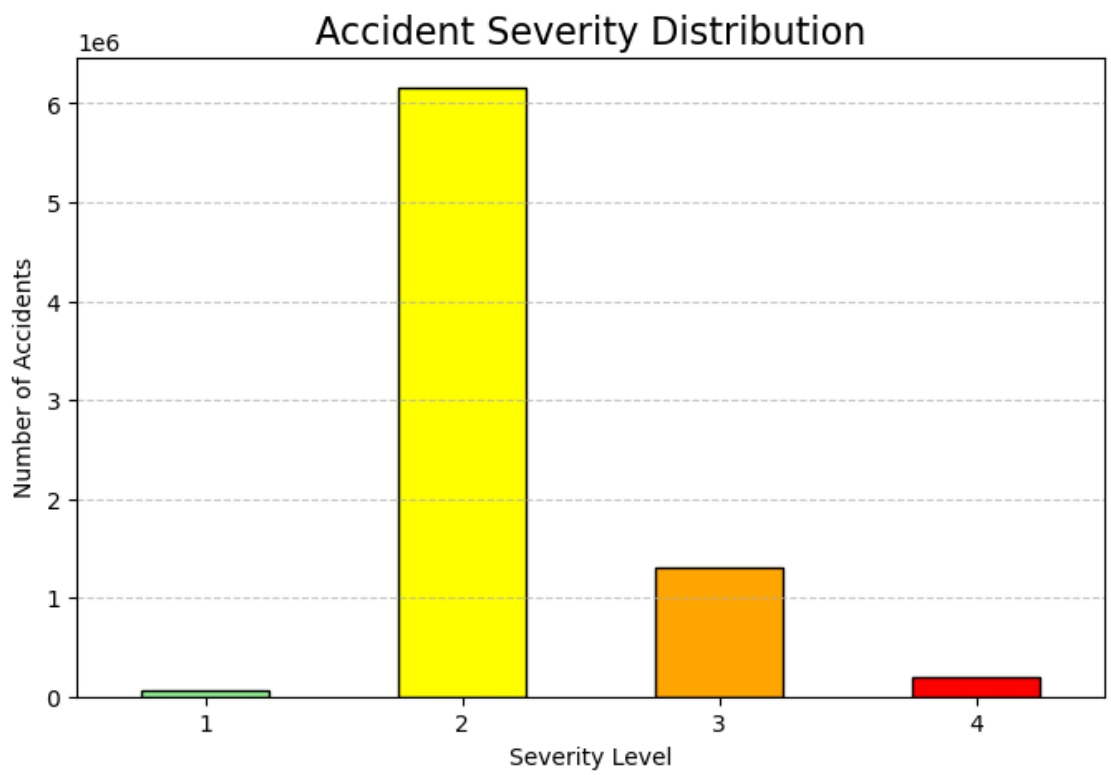


[86]:
```
df['Street'].value_counts().head(10) # Top Dangerous Streets
```

[86]:
```
Street
I-95 N     78430
I-95 S     74528
I-5 N      71968
I-10 E     55572
I-10 W     53725
I-5 S      51781
I-80 W     39662
I-80 E     36113
I-405 N    32364
I-75 N     28166
Name: count, dtype: int64
```

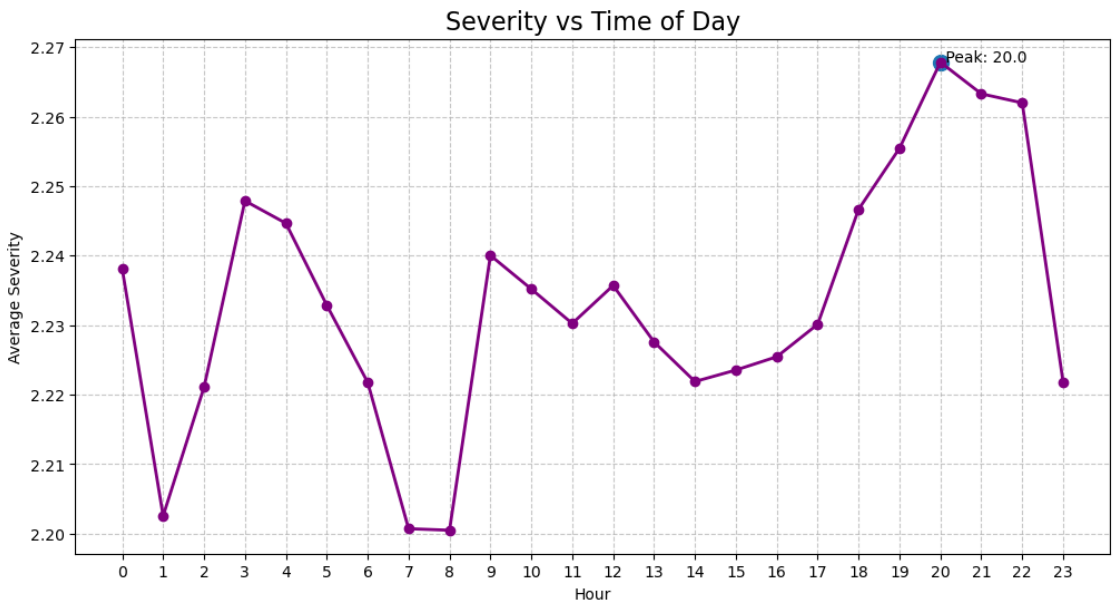
[87]:
```
severity_counts = df['Severity'].value_counts().sort_index()

colors = ['lightgreen', 'yellow', 'orange', 'red'][:len(severity_counts)]

plt.figure(figsize=(8,5))

severity_counts.plot(
    kind='bar',
    color=colors,
    edgecolor='black'
)
```

```
plt.title("Accident Severity Distribution", fontsize=16)
plt.xlabel("Severity Level")
plt.ylabel("Number of Accidents")

plt.xticks(rotation=0)
plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.show()
```



[88]:
```
severity_hour = df.groupby('hour')['Severity'].mean()

plt.figure(figsize=(12,6))

plt.plot(
    severity_hour.index,
    severity_hour.values,
    marker='o',
    linewidth=2,
    color='purple'
)

# Highlight highest severity hour
```

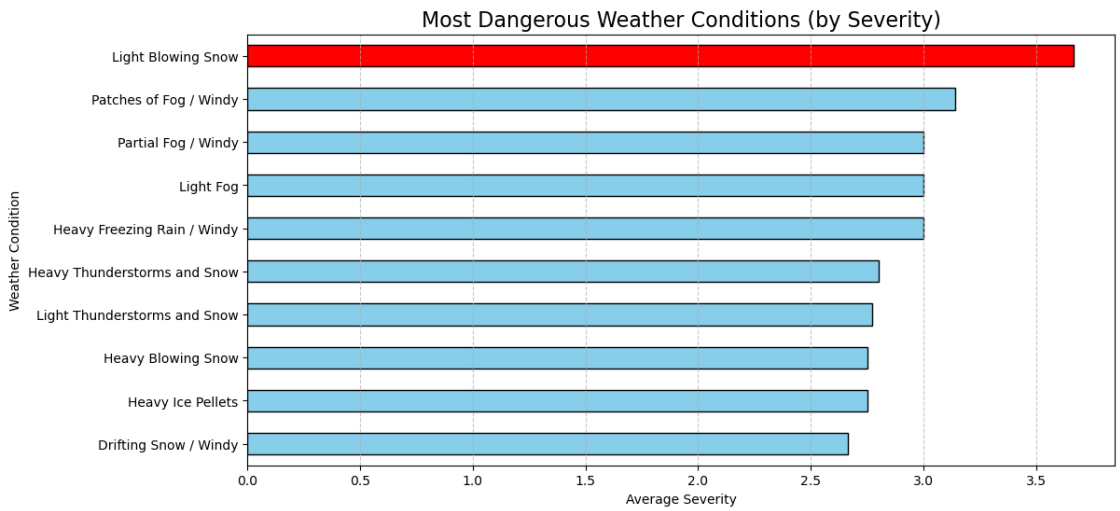
```
max_hour = severity_hour.idxmax()
max_value = severity_hour.max()

plt.scatter(max_hour, max_value, s=100)  # highlight point
plt.text(max_hour, max_value, f' Peak: {max_hour}', fontsize=10)

plt.title("Severity vs Time of Day", fontsize=16)
plt.xlabel("Hour")
plt.ylabel("Average Severity")

plt.xticks(range(0,24))
plt.grid(linestyle='--', alpha=0.7)

plt.show()
```



[90]:
```
weather_severity = df.groupby('Weather_Condition')['Severity'].mean().
↪sort_values().tail(10)
```

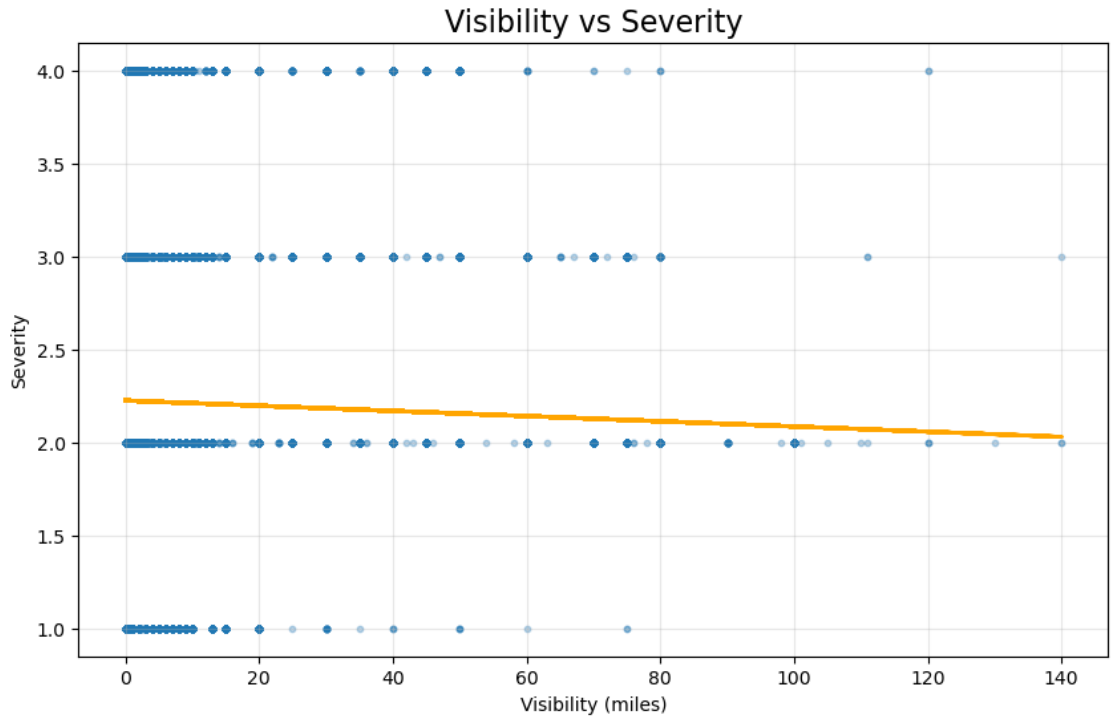
[91]:
```
colors = ['skyblue'] * len(weather_severity)
colors[-1] = 'red'   # highest severity highlighted

plt.figure(figsize=(12,6))

weather_severity.plot(
    kind='barh',
    color=colors,
    edgecolor='black'
```

```
)

plt.title("Most Dangerous Weather Conditions (by Severity)", fontsize=16)
plt.xlabel("Average Severity")
plt.ylabel("Weather Condition")

plt.grid(axis='x', linestyle='--', alpha=0.7)

plt.show()
```



[94]:
```
import numpy as np

plt.figure(figsize=(10,6))

x = df['Visibility(mi)']
y = df['Severity']

plt.scatter(x, y, alpha=0.3, s=10)

# Trend line
z = np.polyfit(x.fillna(0), y.fillna(0), 1)
p = np.poly1d(z)
plt.plot(x, p(x), linewidth=2,color = 'orange')

plt.title("Visibility vs Severity", fontsize=16)
plt.xlabel("Visibility (miles)")
plt.ylabel("Severity")

plt.grid(alpha=0.3)
```

```
plt.show()
```



[95]:
```
df.groupby(pd.cut(df['Precipitation(in)'], bins=5))['Severity'].mean()
↪#Heavy Rain More Accidents
```

[95]:
```
Precipitation(in)
(-0.0365, 7.294]    2.156774
(7.294, 14.588]     2.427711
(14.588, 21.882]         NaN
(21.882, 29.176]    2.166667
(29.176, 36.47]     2.000000
Name: Severity, dtype: float64
```

[96]:
```
df.groupby('Traffic_Signal')['Severity'].mean()     #If signal affects accidents
```

[96]:
```
Traffic_Signal
False    2.233700
True     2.089675
Name: Severity, dtype: float64
```

[97]:
```
features = ['Junction', 'Traffic_Signal', 'Crossing']

for f in features:
    print(df.groupby(f)['Severity'].mean())
```

```
Junction
False    2.205512
True     2.298471
Name: Severity, dtype: float64
Traffic_Signal
False    2.233700
True     2.089675
Name: Severity, dtype: float64
Crossing
False    2.231240
True     2.064467
Name: Severity, dtype: float64
```

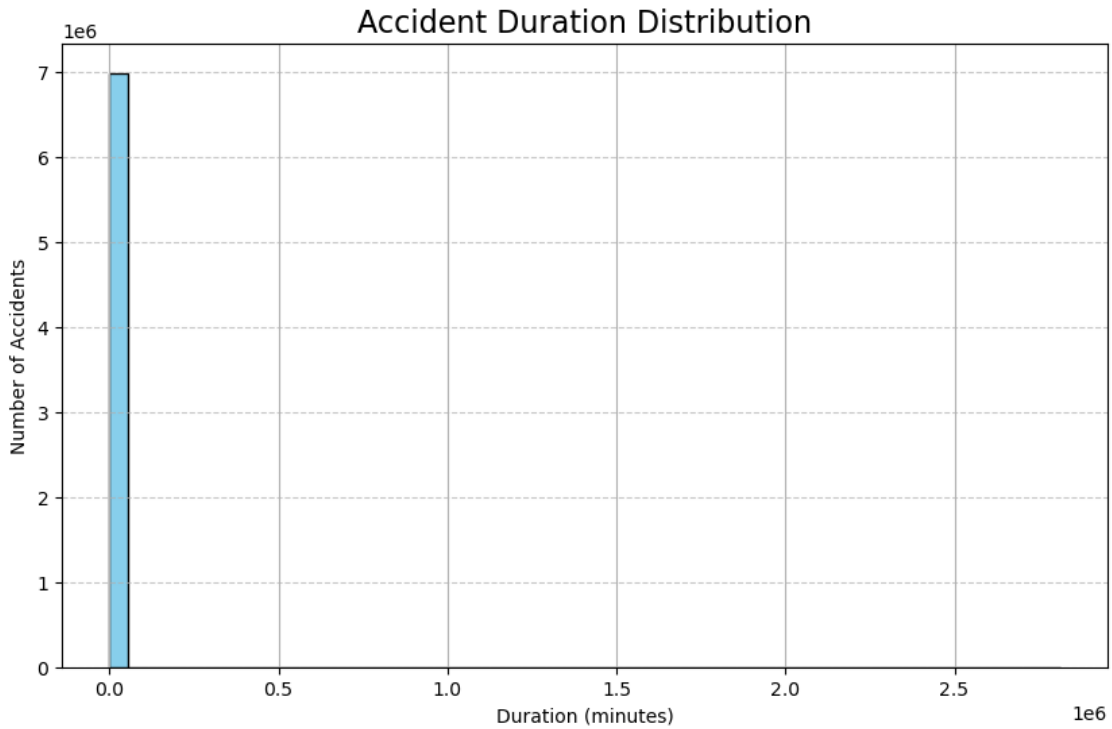
[98]:
```
df['Sunrise_Sunset'].value_counts()
```

[98]:
```
Sunrise_Sunset
Day      5334553
Night    2370595
Name: count, dtype: int64
```

[99]:
```
df.groupby('Sunrise_Sunset')['Severity'].mean()
```

[99]:
```
Sunrise_Sunset
Day      2.209396
Night    2.219069
Name: Severity, dtype: float64
```

[100]:
```
plt.figure(figsize=(10,6))

df['duration'].hist(
    bins=50,
    color='skyblue',
    edgecolor='black'
)

plt.title("Accident Duration Distribution", fontsize=16)
plt.xlabel("Duration (minutes)")
plt.ylabel("Number of Accidents")

plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.show()
```


Accident Duration Distribution
1e6
7
6
5
4
3
2
1
0
Number of Accidents
0.0
0.5
1.0
1.5
2.0
2.5
Duration (minutes)
1e6

[ ]:

```
```